



Figure 1. Two Gaussian density functions where they are equal at the point x^* .

As we have $\mathbb{P}(A, B) = \mathbb{P}(A|B) \mathbb{P}(B)$, we can say:

$$\begin{aligned} \mathbb{P}(\text{error}) &= \mathbb{P}(x > x^* | x \in \mathcal{C}_1) \mathbb{P}(x \in \mathcal{C}_1) \\ &\quad + \mathbb{P}(x < x^* | x \in \mathcal{C}_2) \mathbb{P}(x \in \mathcal{C}_2), \end{aligned} \quad (5)$$

which we want to minimize:

$$\underset{x^*}{\text{minimize}} \quad \mathbb{P}(\text{error}), \quad (6)$$

by finding the best boundary of classes, i.e., x^* .

According to the definition of CDF, we have:

$$\begin{aligned} \mathbb{P}(x < c, x \in \mathcal{C}_1) &= F_1(c), \\ \implies \mathbb{P}(x > x^*, x \in \mathcal{C}_1) &= 1 - F_1(x^*), \end{aligned} \quad (7)$$

$$\mathbb{P}(x < x^*, x \in \mathcal{C}_2) = F_2(x^*). \quad (8)$$

According to the definition of PDF, we have:

$$\mathbb{P}(x \in \mathcal{C}_1) = f_1(x) = \pi_1, \quad (9)$$

$$\mathbb{P}(x \in \mathcal{C}_2) = f_2(x) = \pi_2, \quad (10)$$

where we denote the priors $f_1(x)$ and $f_2(x)$ by π_1 and π_2 , respectively.

Hence, Eqs. (5) and (6) become:

$$\underset{x^*}{\text{minimize}} \quad (1 - F_1(x^*)) \pi_1 + F_2(x^*) \pi_2. \quad (11)$$

We take derivative for the sake of minimization:

$$\begin{aligned} \frac{\partial \mathbb{P}(\text{error})}{\partial x^*} &= -f_1(x^*) \pi_1 + f_2(x^*) \pi_2 \stackrel{\text{set}}{=} 0, \\ \implies f_1(x^*) \pi_1 &= f_2(x^*) \pi_2. \end{aligned} \quad (12)$$

Another way to obtain this expression is equating the posterior probabilities to have the equation of the boundary of classes:

$$\mathbb{P}(x \in \mathcal{C}_1 | X = x) \stackrel{\text{set}}{=} \mathbb{P}(x \in \mathcal{C}_2 | X = x). \quad (13)$$

According to Bayes rule, the *posterior* is:

$$\begin{aligned} \mathbb{P}(x \in \mathcal{C}_1 | X = x) &= \frac{\mathbb{P}(X = x | x \in \mathcal{C}_1) \mathbb{P}(x \in \mathcal{C}_1)}{\mathbb{P}(X = x)} \\ &= \frac{f_1(x) \pi_1}{\sum_{k=1}^{|\mathcal{C}|} \mathbb{P}(X = x | x \in \mathcal{C}_k) \pi_k}, \end{aligned} \quad (14)$$

where $|\mathcal{C}|$ is the number of classes which is two here. The $f_1(x)$ and π_1 are the *likelihood (class conditional)* and *prior* probabilities, respectively, and the denominator is the marginal probability.

Therefore, Eq. (13) becomes:

$$\begin{aligned} &\frac{f_1(x) \pi_1}{\sum_{i=1}^{|\mathcal{C}|} \mathbb{P}(X = x | x \in \mathcal{C}_i) \pi_i} \\ &\stackrel{\text{set}}{=} \frac{f_2(x) \pi_2}{\sum_{i=1}^{|\mathcal{C}|} \mathbb{P}(X = x | x \in \mathcal{C}_i) \pi_i} \\ \implies f_1(x) \pi_1 &= f_2(x) \pi_2. \end{aligned} \quad (15)$$

Now let us think of data as *multivariate* data with dimensionality d . The PDF for multivariate Gaussian distribution, $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is:

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}|}} \exp \left(-\frac{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}{2} \right), \quad (16)$$

where $\mathbf{x} \in \mathbb{R}^d$, $\boldsymbol{\mu} \in \mathbb{R}^d$ is the mean, $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ is the covariance matrix, and $|\cdot|$ is the determinant of matrix. The $\pi \approx 3.14$ in this equation should not be confused with the π_k (prior) in Eq. (12) or (15). Therefore, the Eq. (12) or (15) becomes:

$$\begin{aligned} &\frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}_1|}} \exp \left(-\frac{(\mathbf{x} - \boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}_1^{-1} (\mathbf{x} - \boldsymbol{\mu}_1)}{2} \right) \pi_1 \\ &= \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}_2|}} \exp \left(-\frac{(\mathbf{x} - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_2^{-1} (\mathbf{x} - \boldsymbol{\mu}_2)}{2} \right) \pi_2, \end{aligned} \quad (17)$$

where the distributions of the first and second class are $\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$, respectively.

3. Linear Discriminant Analysis for Binary Classification

In Linear Discriminant Analysis (LDA), we assume that the two classes have equal covariance matrices:

$$\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma}. \quad (18)$$

Therefore, the Eq. (17) becomes:

$$\begin{aligned}
& \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{(x - \mu_1)^\top \Sigma^{-1} (x - \mu_1)}{2}\right) \pi_1 \\
&= \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{(x - \mu_2)^\top \Sigma^{-1} (x - \mu_2)}{2}\right) \pi_2, \\
&\Rightarrow \exp\left(-\frac{(x - \mu_1)^\top \Sigma^{-1} (x - \mu_1)}{2}\right) \pi_1 \\
&= \exp\left(-\frac{(x - \mu_2)^\top \Sigma^{-1} (x - \mu_2)}{2}\right) \pi_2, \\
&\stackrel{(a)}{\Rightarrow} -\frac{1}{2}(x - \mu_1)^\top \Sigma^{-1} (x - \mu_1) + \ln(\pi_1) \\
&= -\frac{1}{2}(x - \mu_2)^\top \Sigma^{-1} (x - \mu_2) + \ln(\pi_2),
\end{aligned}$$

where (a) takes natural logarithm from the sides of equation.

We can simplify this term as:

$$\begin{aligned}
& (x - \mu_1)^\top \Sigma^{-1} (x - \mu_1) = (x^\top - \mu_1^\top) \Sigma^{-1} (x - \mu_1) \\
&= x^\top \Sigma^{-1} x - x^\top \Sigma^{-1} \mu_1 - \mu_1^\top \Sigma^{-1} x + \mu_1^\top \Sigma^{-1} \mu_1 \\
&\stackrel{(a)}{=} x^\top \Sigma^{-1} x + \mu_1^\top \Sigma^{-1} \mu_1 - 2 \mu_1^\top \Sigma^{-1} x,
\end{aligned} \quad (19)$$

where (a) is because $x^\top \Sigma^{-1} \mu_1 = \mu_1^\top \Sigma^{-1} x$ as it is a scalar and Σ^{-1} is symmetric so $\Sigma^{-\top} = \Sigma^{-1}$. Thus, we have:

$$\begin{aligned}
& -\frac{1}{2} x^\top \Sigma^{-1} x - \frac{1}{2} \mu_1^\top \Sigma^{-1} \mu_1 + \mu_1^\top \Sigma^{-1} x + \ln(\pi_1) \\
&= -\frac{1}{2} x^\top \Sigma^{-1} x - \frac{1}{2} \mu_2^\top \Sigma^{-1} \mu_2 + \mu_2^\top \Sigma^{-1} x + \ln(\pi_2).
\end{aligned}$$

Therefore, if we multiply the sides of equation by 2, we have:

$$\begin{aligned}
& 2 (\Sigma^{-1} (\mu_2 - \mu_1))^\top x \\
&+ (\mu_1 - \mu_2)^\top \Sigma^{-1} (\mu_1 - \mu_2) + 2 \ln\left(\frac{\pi_2}{\pi_1}\right) = 0,
\end{aligned} \quad (20)$$

which is the equation of a line in the form of $a^\top x + b = 0$. Therefore, if we consider Gaussian distributions for the two classes where the covariance matrices are assumed to be equal, the decision boundary of classification is a line. Because of linearity of the decision boundary which discriminates the two classes, this method is named *linear discriminant analysis*.

For obtaining Eq. (20), we brought the expressions to the right side which was corresponding to the second class; therefore, if we use $\delta(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ as the left-hand-side expression (function) in Eq. (20):

$$\begin{aligned}
\delta(x) &:= 2 (\Sigma^{-1} (\mu_2 - \mu_1))^\top x \\
&+ (\mu_1 - \mu_2)^\top \Sigma^{-1} (\mu_1 - \mu_2) + 2 \ln\left(\frac{\pi_2}{\pi_1}\right),
\end{aligned} \quad (21)$$

the class of an instance x is estimated as:

$$\hat{C}(x) = \begin{cases} 1, & \text{if } \delta(x) < 0, \\ 2, & \text{if } \delta(x) > 0. \end{cases} \quad (22)$$

If the priors of two classes are equal, i.e., $\pi_1 = \pi_2$, the Eq. (20) becomes:

$$\begin{aligned}
& 2 (\Sigma^{-1} (\mu_2 - \mu_1))^\top x \\
&+ (\mu_1 - \mu_2)^\top \Sigma^{-1} (\mu_1 - \mu_2) = 0,
\end{aligned} \quad (23)$$

whose left-hand-side expression can be considered as $\delta(x)$ in Eq. (22).

4. Quadratic Discriminant Analysis for Binary Classification

In Quadratic Discriminant Analysis (QDA), we relax the assumption of equality of the covariance matrices:

$$\Sigma_1 \neq \Sigma_2, \quad (24)$$

which means the covariances are not *necessarily* equal (if they are actually equal, the decision boundary will be linear and QDA reduces to LDA).

Therefore, the Eq. (17) becomes:

$$\begin{aligned}
& \frac{1}{\sqrt{(2\pi)^d |\Sigma_1|}} \exp\left(-\frac{(x - \mu_1)^\top \Sigma_1^{-1} (x - \mu_1)}{2}\right) \pi_1 \\
&= \frac{1}{\sqrt{(2\pi)^d |\Sigma_2|}} \exp\left(-\frac{(x - \mu_2)^\top \Sigma_2^{-1} (x - \mu_2)}{2}\right) \pi_2, \\
&\stackrel{(a)}{\Rightarrow} -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma_1|) \\
&\quad - \frac{1}{2} (x - \mu_1)^\top \Sigma_1^{-1} (x - \mu_1) + \ln(\pi_1) \\
&= -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma_2|) \\
&\quad - \frac{1}{2} (x - \mu_2)^\top \Sigma_2^{-1} (x - \mu_2) + \ln(\pi_2),
\end{aligned}$$

where (a) takes natural logarithm from the sides of equation. According to Eq. (19), we have:

$$\begin{aligned}
& -\frac{1}{2} \ln(|\Sigma_1|) - \frac{1}{2} x^\top \Sigma_1^{-1} x - \frac{1}{2} \mu_1^\top \Sigma_1^{-1} \mu_1 \\
&+ \mu_1^\top \Sigma_1^{-1} x + \ln(\pi_1) \\
&= -\frac{1}{2} \ln(|\Sigma_2|) - \frac{1}{2} x^\top \Sigma_2^{-1} x - \frac{1}{2} \mu_2^\top \Sigma_2^{-1} \mu_2 \\
&+ \mu_2^\top \Sigma_2^{-1} x + \ln(\pi_2).
\end{aligned}$$

Therefore, if we multiply the sides of equation by 2, we have:

$$\begin{aligned}
& x^\top (\Sigma_1 - \Sigma_2)^{-1} x + 2 (\Sigma_2^{-1} \mu_2 - \Sigma_1^{-1} \mu_1)^\top x \\
&+ (\mu_1^\top \Sigma_1^{-1} \mu_1 - \mu_2^\top \Sigma_2^{-1} \mu_2) + \ln\left(\frac{|\Sigma_1|}{|\Sigma_2|}\right) \\
&+ 2 \ln\left(\frac{\pi_2}{\pi_1}\right) = 0,
\end{aligned} \quad (25)$$